Jiyoon Shin

(470) 883-6604 | jyoon.shin12@gmail.com | [LinkedIn] | [Portfolio]

EDUCATION

Georgia Institute of Technology — Atlanta, GA
M.S. in Computer Science (Machine Learning) — May 2028

Yonsei University — South Korea
B.S. in Urban Planning and Engineering & Applied Statistics (GPA: 3.9/4.0) — Feb. 2026

EXPERIENCE

Smart City Infrastructure Lab, Georgia Institute of Technology — Aug. 2026 - Present
Graduate Research Assistant — Atlanta, GA

- Conducted research combining advanced sensing, AI, and geospatial analytics for transportation safety and mobility
- Modeled crash frequency at high-stress intersections using Negative Binomial regression across 2,200+ intersections
- Established a screening framework to prioritize high-risk intersections and guide targeted safety improvements

Kearney — Mar. 2026 - Jun. 2026
Research Analyst — South Korea

- Built and maintained ETL pipelines to process 40M+ traditional and social media records for PR performance reporting
- Automated dashboard validation against PostgreSQL records using AWS Lambda to keep dashboards up to date
- Engineered an LLM-based classification pipeline for 355K+ multilingual social posts, classifying relevance, category, and sentiment via batch processing with ~89% accuracy on sampled content
- Developed a multilingual newsroom tracking system to crawl and match local articles with HQ releases across 40+ countries using BERT and CLIP embeddings, cutting review time by half

Kakao Mobility — May 2025 - Feb. 2026
Data Science Intern — South Korea

- Conducted root cause analysis of ETA errors at the road-link level, uncovering patterns to guide model improvements
- Redefined route adherence using distance weighting; deployed via scheduled SQL and a monitoring dashboard
- Estimated traffic signal cycles from GPS data using Lomb-Scargle and autocorrelation to infer reliable signal timing
- Proposed Transformer-based imputation to improve pattern-speed estimation when historical data were sparse
- Addressed limitations of static road types in capturing driving-pattern dynamics by reclassifying roads using PCA and K-means, achieving ARI 0.85 across dates

Transportation Planning Lab, Yonsei University — Mar. 2024 - Jun. 2024
Undergraduate Research Assistant — South Korea

- Analyzed 15K+ trips using t-tests, ANOVA, and stepwise regression to identify drivers of railway station access time

PROJECTS

Seoul Digital Connectivity Analysis | Special Recognition, ITU UMC Data Hackathon — Dec. 2025 - Jun. 2026

- Modeled digital usage across individual and district levels using a hierarchical linear model on 10K+ survey responses
- Combined 130K+ LLM-classified community posts with administrative data using Bayesian updating, surfacing digital-connectivity signals not captured by official indicators

Impact of Demand-Responsive Transit (DRT) on Property Values | First Place, Yonsei University — Sep. 2025 - Dec. 2025

- Estimated property-value effects of DRT using weighted regression discontinuity around launch dates, identifying heterogeneous effects across market conditions

SKILLS

Languages: Python, SQL, R, Java
Data & Platforms: BigQuery, PostgreSQL, Airflow, GCP, AWS, Docker, Tableau, Git
Statistics & ML: Statistical Inference, Bayesian Modeling, Predictive Modeling, Time-Series Analysis, Transformers

PUBLICATIONS

- Ziyu Liu, ..., **Jiyoon Shin**, et al. "AD-GPT: Large Language Models in Alzheimer's Disease." *BMC Medical Informatics and Decision Making* (2026). https://doi.org/10.1186/s12911-026-03579-x